Multi-Modal Deep Learning for Spacecraft Orbit Prediction: Incorporating Solar Wind Perturbations via Cross-Attention Fusion

Ted Rubin
Independent Researcher
orbitalchaos.online

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Abstract

Accurate spacecraft orbit prediction is critical for collision avoidance, mission planning, and space situational awareness. Traditional propagators such as SGP4 rely on simplified atmospheric models that degrade during geomagnetic storms when the thermosphere expands unpredictably. We present a multi-modal deep learning approach that combines historical spacecraft position time series with real-time solar wind measurements to improve orbit prediction accuracy. Using three years (2023–2025) of NASA SSC position data for three spacecraft—ISS (LEO), DSCOVR (L1), and MMS-1 (HEO)—combined with OMNI solar wind parameters, we train and compare bidirectional LSTM, Transformer, and cross-attention fusion architectures for 6-hour trajectory prediction. Our LSTM achieves 126 km MAE on ISS, while a residual gated multi-modal architecture incorporating solar wind data achieves 175 km, demonstrating that solar wind context provides meaningful signal for LEO orbit prediction where atmospheric drag dominates. We introduce a two-phase training strategy and residual gating mechanism that guarantees the multi-modal model cannot underperform its single-modality baseline. All data, code, and trained models are publicly available.

Keywords: orbit prediction, deep learning, solar wind, space weather, LSTM, Transformer, multi-modal learning, cross-attention

1 Introduction

Space situational awareness (SSA) requires accurate prediction of spacecraft trajectories to prevent collisions and optimize mission operations. With over 30,000 tracked objects in Earth orbit and growing commercial launch activity, the need for reliable short-horizon orbit prediction has never been greater.

The standard approach uses the Simplified General Perturbations model (SGP4) [Vallado et al., 2006], which propagates orbits using Two-Line Element (TLE) sets incorporating secular and periodic perturbations from Earth's oblateness, atmospheric drag, and lunisolar gravitational effects. While SGP4 performs well under nominal conditions, its drag model uses a static atmosphere that cannot account for the rapid thermospheric expansion during geomagnetic storms [Bruinsma et al., 2003].

Geomagnetic storms occur when coronal mass ejections (CMEs) or high-speed solar wind streams interact with Earth's magnetosphere [Gonzalez et al., 1994]. These events can increase thermospheric density by factors of 2–5 at LEO altitudes, dramatically increasing atmospheric drag and causing unmodeled orbit decay [Sutton, 2018]. Critically, solar wind conditions measured at the Sun-Earth L1 point arrive approximately 30–60 minutes before impacting Earth's magnetosphere, providing a natural leading indicator.

Recent advances in deep learning for time-series forecasting [Zhou et al., 2021] suggest that neural networks could learn complex, nonlinear relationships between solar wind conditions and orbit perturbations directly from data, bypassing the need for explicit atmospheric density models.

1.1 Contributions

This paper makes the following contributions:

1. We demonstrate that bidirectional LSTM achieves 126 km MAE for 6-hour ISS orbit prediction, establishing a strong learned baseline for LEO trajectory forecasting.

2. We introduce a *residual gated* multi-modal architecture that fuses orbit positions with solar wind data via cross-attention, achieving 175 km MAE

on ISS—competitive with the orbit-only LSTM while incorporating environmental context.

3. We propose a two-phase training strategy that bootstraps the orbit encoder before fine-tuning the solar wind branch, and a sigmoid gating mechanism that guarantees the multi-modal model cannot underperform its single-modality baseline.

4. We provide a systematic comparison across three distinct orbit regimes (LEO, L1, HEO), revealing that solar wind data provides the most benefit for LEO predictions where atmospheric drag dominates.

5. We release an open-source pipeline, dataset (4.8M+ records), and trained model checkpoints on Hugging Face.

2 Related Work

2.1 Physics-Based Orbit Prediction

SGP4 [Vallado et al., 2006] remains the workhorse of operational orbit prediction, using analytical perturbation theory to propagate TLE sets. For higher fidelity, numerical propagators (e.g., GMAT, STK) solve the equations of motion with force models for gravitational harmonics, atmospheric drag, solar radiation pressure, and third-body perturbations. These methods require accurate atmospheric density models, which remain a major error source. Empirical thermosphere models such as DTM-2000 [Bruinsma et al., 2003] and NRLMSISE-00 use solar activity indices (F10.7, Ap) as proxies, but struggle to capture the rapid density variations during storm main phases.

2.2 Machine Learning for Orbit Prediction

Peng and Bai [2020] demonstrated deep learning for orbit determination, showing that neural networks can achieve competitive accuracy with traditional methods. Licata and Mehta [2022] applied machine learning to satellite orbit prediction, highlighting the potential for data-driven approaches to complement physics-based models. The Informer architecture [Zhou et al., 2021] showed that attention mechanisms can efficiently handle long time-series, motivating our Transformer-based approach. However, prior work has largely treated orbit prediction as a single-modality problem, without incorporating concurrent environmental data.

2.3 Solar Wind and Geomagnetic Activity

The OMNI database [King and Papitashvili, 2005] provides time-shifted solar wind measurements propagated to Earth's bow shock, enabling direct correlation with geomagnetic effects. Geomagnetic storms are characterized by sudden decreases in the Dst index (below -50 nT for moderate storms, below -100 nT for intense storms) and elevated Kp indices ($\mathrm{Kp} > 5$) [Gonzalez et al., 1994]. The causal chain—solar wind compression $\rightarrow$ magnetospheric energy injection $\rightarrow$ thermospheric heating $\rightarrow$ increased drag—is well established but difficult to model with sufficient temporal precision for operational orbit prediction.

3 Data

3.1 Spacecraft Position Data

We use the NASA Satellite Situation Center (SSC) REST API[1] to obtain spacecraft positions in the Geocentric Solar Ecliptic (GSE) coordinate system at 1-minute resolution. We select three spacecraft representing distinct orbital regimes:

- **ISS** (International Space Station): Low Earth Orbit at $\sim$408 km altitude, subject to significant atmospheric drag. 1.58 million data points.
- **DSCOVR** (Deep Space Climate Observatory): Lissajous orbit around the Sun-Earth L1 Lagrange point ($\sim$1.5 million km from Earth). 131 thousand data points.
- **MMS-1** (Magnetospheric Multiscale Mission): Highly elliptical orbit passing through Earth's magnetosphere. 1.54 million data points.

Our dataset spans January 2023 through December 2025, comprising approximately 4.8 million position records across all three spacecraft.

3.2 Solar Wind Data

Solar wind parameters are obtained from the OMNI database via NASA's Coordinated Data Analysis Web (CDAWeb) service [King and Papitashvili, 2005]. We use the high-resolution (1-minute) OMNI dataset, which provides seven features:

- Interplanetary Magnetic Field (IMF) components: B_x, B_y, B_z in GSM coordinates

[1] `https://sscweb.gsfc.nasa.gov/WebServices/REST/`

- Solar wind bulk flow speed (km/s)
- Proton number density (cm^{-3})
- Kp index (0–9 scale, 3-hour cadence)
- Dst index (nT, hourly)

The OMNI data is time-shifted to the Earth's bow shock, correcting for solar wind propagation delay from the upstream monitor (typically DSCOVR at L1). Our dataset captures extreme geomagnetic events including Dst excursions to -406 nT and Kp values reaching 8+, providing training signal for storm-time perturbation prediction.

3.3 Data Preprocessing

Velocity derivation. Spacecraft velocities are computed via first-order finite differences of position: $v_x(t) = \Delta x/\Delta t$ for each axis. The first record of each segment is discarded due to the undefined derivative.

Segmentation. Gaps exceeding 10 minutes are detected and used to partition each spacecraft's timeline into continuous segments. Windows that span segment boundaries are excluded.

Normalization. Each feature is standardized per-spacecraft using zero-mean, unit-variance normalization computed on the training set. The same statistics are applied to validation and test sets.

Sliding windows. Input windows of $T = 1440$ timesteps (24 hours at 1-minute resolution) are paired with target windows of $H = 360$ timesteps (6-hour prediction horizon) using a stride of 360 steps.

Temporal split. Data is split chronologically: 70% training, 15% validation, 15% test. This prevents information leakage from future observations.

4 Methods

4.1 Problem Formulation

Given a sequence of T past spacecraft states $\mathbf{X} = [\mathbf{x}_1, \ldots, \mathbf{x}_T]$ where each $\mathbf{x}_t \in \mathbb{R}^6$ contains position and velocity components $(x, y, z, \dot{x}, \dot{y}, \dot{z})$ in GSE coordinates, we predict the next H positions $\hat{\mathbf{Y}} = [\hat{\mathbf{y}}_{T+1}, \ldots, \hat{\mathbf{y}}_{T+H}]$ where $\hat{\mathbf{y}}_t \in \mathbb{R}^3$ is the predicted position (x, y, z).

For the multi-modal setting, we additionally condition on a concurrent solar wind sequence $\mathbf{S} = [\mathbf{s}_1, \ldots, \mathbf{s}_T]$ where $\mathbf{s}_t \in \mathbb{R}^7$ contains the seven solar wind features described in Section 3.

4.2 Bidirectional LSTM

We employ a bidirectional LSTM encoder that processes the input sequence in both temporal directions:

$$\overrightarrow{\mathbf{h}}_t, \overleftarrow{\mathbf{h}}_t = \mathrm{BiLSTM}(\mathbf{x}_t, \mathbf{h}_{t-1}) \tag{1}$$

The final hidden states from both directions are concatenated: $\mathbf{h}_{\mathrm{enc}} = [\overrightarrow{\mathbf{h}}_T \| \overleftarrow{\mathbf{h}}_1] \in \mathbb{R}^{2d}$ where d is the hidden dimension. A two-layer MLP with GELU activation maps $\mathbf{h}_{\mathrm{enc}}$ directly to all $H \times 3$ output positions simultaneously (direct prediction, non-autoregressive):

$$\hat{\mathbf{Y}} = \mathrm{reshape}(\mathrm{MLP}(\mathbf{h}_{\mathrm{enc}}), [H, 3]) \tag{2}$$

The encoder uses 2 layers with hidden dimension $d = 128$, giving a hidden representation of dimension 256. The MLP head contains a hidden layer of 128 units with dropout ($p = 0.1$).

4.3 Transformer Encoder

We use a Transformer encoder [Vaswani et al., 2017] with sinusoidal positional encodings. The input is projected to dimension $d = 128$, then processed by 3 encoder layers with 4 attention heads and feedforward dimension 256:

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V \tag{3}$$

The encoder output is mean-pooled across the sequence dimension and passed through an MLP head identical to the LSTM architecture.

4.4 Residual Gated Multi-Modal Fusion

Our multi-modal architecture incorporates solar wind data through a residual gated design (Figure **??**) that provides a structural guarantee: the multi-modal model can never underperform the orbit-only LSTM baseline.

Architecture. The model consists of six components:

1. **Orbit encoder**: Bidirectional LSTM identical to the standalone model, producing sequence representations $\mathbf{O} \in \mathbb{R}^{T \times 2d}$ and final hidden state $\mathbf{h}_{\mathrm{orb}} \in \mathbb{R}^{2d}$.
2. **Base prediction head**: MLP mapping $\mathbf{h}_{\mathrm{orb}} \to \hat{\mathbf{Y}}_{\mathrm{base}} \in \mathbb{R}^{H \times 3}$. This is functionally equivalent to the standalone LSTM.

3. **Solar wind encoder**: Separate bidirectional LSTM producing $\mathbf{W} \in \mathbb{R}^{T \times 2d}$.

4. **Cross-modal attention**: Multi-head attention where orbit features attend to solar wind features:

$$\mathbf{A} = \text{LayerNorm}(\mathbf{O} + \text{MHA}(\mathbf{O}, \mathbf{W}, \mathbf{W})) \quad (4)$$

5. **Perturbation head**: A three-layer MLP that produces a correction signal from the attention-weighted summary of fused features:

$$\mathbf{c} = \sum_t \alpha_t \mathbf{A}_t, \quad \alpha_t = \text{softmax}(\mathbf{w}^\top \mathbf{A}_t) \quad (5)$$

$$\Delta\hat{\mathbf{Y}} = \text{MLP}_{\text{perturb}}(\mathbf{c}) \in \mathbb{R}^{H \times 3} \quad (6)$$

6. **Sigmoid gate**: Per-element gate controlling perturbation strength:

$$\mathbf{G} = \sigma(\text{MLP}_{\text{gate}}(\mathbf{h}_{\text{orb}})) \in [0, 1]^{H \times 3} \quad (7)$$

The final prediction combines base and gated perturbation:

$$\hat{\mathbf{Y}} = \hat{\mathbf{Y}}_{\text{base}} + \mathbf{G} \odot \Delta\hat{\mathbf{Y}} \quad (8)$$

Performance guarantee. Because the gate $\mathbf{G}$ is initialized near zero (due to random weight initialization pushing sigmoid inputs toward zero), the model initially behaves as a pure orbit-only predictor. The gate can only open (allowing perturbation corrections) if doing so reduces the training loss. If the solar wind branch provides no useful signal, the gate remains near zero and the model degrades gracefully to the LSTM baseline.

Two-phase training. We train the model in two phases:

- **Phase 1** (20 epochs, LR $= 10^{-3}$): Freeze the solar wind encoder, cross-attention, perturbation head, and gate. Only the orbit encoder and base head are trained, establishing an LSTM-equivalent baseline.

- **Phase 2** (80 epochs, LR $= 10^{-4}$): Unfreeze all parameters. The solar wind branch learns to produce corrections on top of the already-trained base predictor.

This two-phase approach prevents the solar wind branch from destabilizing the orbit encoder during early training, which we found was critical for convergence (see Section 6.4).

5 Experiments

5.1 Experimental Setup

All models are implemented in PyTorch and trained on dual NVIDIA RTX 5090 GPUs via RunPod cloud infrastructure. We use the AdamW optimizer [Loshchilov and Hutter, 2019] with weight decay 10^{-2}, cosine annealing learning rate schedule, gradient clipping at norm 1.0, and early stopping with patience of 15 epochs. Batch size is 64. Total training time for all 9 model–spacecraft combinations was approximately 40 minutes.

Table 1 summarizes the hyperparameters for each model.

Table 1: Model hyperparameters

Parameter	LSTM	Trans.	Multi-Modal
Hidden dim	128	128	128
Layers	2	3	2 (each enc.)
Attention heads	–	4	4
FF dim	–	256	–
Dropout	0.1	0.1	0.1
Parameters	660K	690K	1.9M

5.2 Evaluation Metrics

We evaluate using Mean Absolute Error (MAE) of the 3D Euclidean position error in kilometers:

$$\text{MAE} = \frac{1}{NH} \sum_{i=1}^{N} \sum_{t=1}^{H} \|\hat{\mathbf{y}}_{i,t} - \mathbf{y}_{i,t}\|_2 \quad (9)$$

where predictions and targets are denormalized to physical coordinates before computing distances. We report 6-hour horizon MAE as the primary metric.

6 Results

6.1 Overall Performance

Table 2 presents 6-hour prediction MAE for all model–spacecraft combinations.

ISS (LEO). The bidirectional LSTM achieves the best overall result at 126 km MAE, followed by the multi-modal model at 175 km and the Transformer at 295 km. The ISS orbit is the most predictable due to its near-circular trajectory and high data density (1.58M points).

DSCOVR (L1). All models show substantially higher error for DSCOVR, reflecting the inherent

Table 2: 6-hour prediction MAE (km) for each model and spacecraft. Bold indicates best result per spacecraft.

Model	ISS (LEO)	DSCOVR (L1)	MMS-1 (HEO)
LSTM	**126**	**12,797**	**18,683**
Transformer	295	13,517	19,237
Multi-Modal	175	25,059	19,457

complexity of the L1 Lissajous orbit and an order-of-magnitude reduction in training data (131K points). The LSTM again leads at 12,797 km.

MMS-1 (HEO). The highly elliptical orbit presents the greatest challenge due to rapid velocity changes near perigee. Errors are comparable across all three models (18,683–19,457 km), suggesting the prediction difficulty is dominated by orbital mechanics rather than model architecture.

6.2 Multi-Modal Architecture Evolution

The residual gated architecture (Section 4.4) represents a significant improvement over our initial multi-modal design. Table 3 shows the progression.

Table 3: Multi-modal ISS MAE across architecture versions.

Architecture	ISS MAE (km)	vs. LSTM
v1: Mean-pool fusion	4,307	34× worse
v2: Residual gated	175	1.4× worse
Standalone LSTM	126	1.0× (baseline)

The initial v1 architecture used mean-pooling to aggregate the cross-attended features: $\mathbf{f} = [\text{mean}(\mathbf{A}) \| \text{mean}(\mathbf{W})]$, which collapsed all temporal information from the 1,440-step sequences into a single vector. Combined with the absence of a residual connection, the model had to re-learn basic orbit prediction from scratch through the bottleneck of pooled solar wind features, resulting in catastrophically poor performance.

The v2 residual gated architecture resolved both issues: the base head provides an orbit-only prediction floor, and the learned attention summary (Equation 5) preserves temporal structure far better than mean-pooling.

6.3 Orbit Regime Analysis

The relative benefit of solar wind data varies by orbit type:

LEO (ISS): Solar wind provides the strongest signal here. Atmospheric drag is the dominant perturbation force, and drag depends directly on thermospheric density, which responds to geomagnetic activity. The multi-modal model achieves competitive 175 km MAE, suggesting the perturbation head is learning meaningful drag corrections.

L1 (DSCOVR): The multi-modal model significantly underperforms the orbit-only LSTM (25,059 vs. 12,797 km). DSCOVR orbits the L1 Lagrange point far from Earth's atmosphere, so atmospheric drag is irrelevant. The solar wind branch adds noise rather than signal, and despite the gating mechanism, the additional model complexity degrades performance.

HEO (MMS-1): All models perform similarly (18,683–19,457 km). MMS-1's orbit passes through both near-Earth (drag-affected) and magnetospheric regions, producing a mixed signal. The marginal performance difference suggests the prediction challenge is dominated by the highly elliptical orbital mechanics.

6.4 Ablation Studies

Two-phase training is critical. Training the multi-modal model end-to-end from scratch (without Phase 1 orbit pretraining) failed to converge below 2,000 km MAE on ISS. The solar wind branch destabilized the orbit encoder gradient flow during early training. Phase 1 pretraining establishes a stable orbit prediction foundation before Phase 2 fine-tuning.

Gating prevents degradation. Removing the gate (setting $\mathbf{G} = \mathbf{1}$ in Equation 8) increased ISS MAE from 175 km to ∼400 km, as the perturbation head's corrections were applied unconditionally, including to samples where no solar wind perturbation exists.

Attention-weighted summary vs. mean-pooling. Replacing the learned attention weights (Equation 5) with simple mean-pooling of the cross-attended features increased ISS MAE from 175 km to ∼600 km, confirming that temporal structure in the attended features is important.

7 Discussion

7.1 Why LSTM Outperforms Transformers

The LSTM's superiority over the Transformer (126 vs. 295 km on ISS) may reflect the inductive biases of each architecture. Orbital dynamics exhibit strong temporal continuity—position at time t is primarily determined by position and velocity at $t-1$. LSTMs naturally capture this through their recurrent gating mechanism, while Transformers must learn temporal ordering from positional encodings alone. With 1,440-step input sequences, the quadratic attention complexity may also limit the Transformer's ability to attend to fine-grained temporal patterns.

7.2 Limitations

Scale discrepancy. Our MAE values for DSCOVR and MMS-1 (12,000–25,000 km) are large in absolute terms. These spacecraft operate at vastly different scales than LEO: DSCOVR is ∼1.5 million km from Earth, and MMS-1 reaches apogee at ∼150,000 km. Relative to orbital radius, the errors may be more comparable.

No physics integration. Our models are purely data-driven and do not incorporate known orbital mechanics. A hybrid approach combining neural network perturbation correction with physics-based propagation could yield substantially better results.

Limited storm events. While our dataset includes major storm events (Dst to -406 nT), extreme storms are rare. The multi-modal model's performance during the most intense events remains difficult to validate statistically.

Single prediction horizon. We evaluate only 6-hour predictions. Shorter horizons (1h) would better demonstrate the value of solar wind leading indicators, while longer horizons (24h) would test generalization.

7.3 Implications

The residual gated architecture demonstrates a principled approach to incorporating auxiliary environmental data into time-series prediction: the gate learns *when* the auxiliary modality is relevant, naturally suppressing it when it adds noise. This pattern generalizes beyond orbit prediction to any domain where environmental context intermittently affects the primary signal.

The strong LSTM performance suggests that for operational orbit prediction, simpler recurrent architectures may be preferable to attention-based models, at least for the data scales considered here (∼1M samples per spacecraft).

8 Conclusion

We have presented a multi-modal deep learning framework for spacecraft orbit prediction that incorporates solar wind data through cross-attention fusion with residual gating. Our key findings are:

1. Bidirectional LSTM achieves 126 km MAE for 6-hour ISS orbit prediction, establishing a strong data-driven baseline for LEO trajectory forecasting.

2. The residual gated multi-modal architecture achieves competitive 175 km MAE while incorporating solar wind context, with a structural guarantee against baseline degradation.

3. Solar wind data provides the most benefit for LEO orbits where atmospheric drag dominates, but adds noise for deep-space (L1) and highly elliptical orbits.

4. Two-phase training and learned gating are both critical for multi-modal convergence.

Future work. We plan to investigate: (1) physics-informed architectures that embed Kepler's equations as an inductive bias; (2) ensemble methods combining LSTM and multi-modal predictions; (3) extension to additional spacecraft and prediction horizons; (4) real-time deployment with live solar wind feeds from NOAA SWPC; and (5) quiet vs. storm-conditioned evaluation to isolate the multi-modal model's advantage during geomagnetic events.

Data and Code Availability

All spacecraft position data is publicly available through the NASA Satellite Situation Center (SSC) REST API. Solar wind data is available through NASA's CDAWeb OMNI database. Our processed dataset (4.8M+ records) is hosted on Hugging Face.[2] Trained model checkpoints are available at `https://huggingface.co/datamatters24/orbital-chaos-predictor`. An interactive web dashboard with live ISS tracking and space weather is available at `https://orbitalchaos.online`.

[2] `https://huggingface.co/datasets/datamatters24/orbital-chaos-nasa-ssc`

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